CCNA 200-301 V2.0

08 VLANs and Access Ports

Part II — Switching and Network Access

EXAM TOPICS

2.2

LECTURER

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THE COURSE

Where this chapter sits

What you will be able to do

- **Explain** what a VLAN changes about a switch, in terms of the learning-and-flooding algorithm from Chapter 3.
- **Create** VLANs and assign access ports to them.
- **Verify** VLAN membership from three different angles, and know which one to trust when they disagree.
- **Diagnose** the classic VLAN faults: a port in the wrong VLAN, a VLAN that does not exist, and a VLAN that is shut down.

ASSUMED KNOWLEDGE

Before we start

Chapter 3 for the MAC table and flooding; Chapter 7 for moving around the command line.

KEY TERMS

Vocabulary for this chapter

- VLAN
- broadcast domain
- access port
- VLAN database
- default VLAN
- native VLAN
- management VLAN
- SVI
- VLAN 1
- shutdown VLAN

Each is defined where it first matters — not here.

What a VLAN actually does

DEFINITION

VLAN

A **VLAN** is a broadcast domain implemented in software on a switch. Ports in VLAN 10 behave as though they were on a physically separate switch from ports in VLAN 20: a broadcast from one never reaches the other, and a frame cannot cross between them without passing through something that routes.

CONCEPT

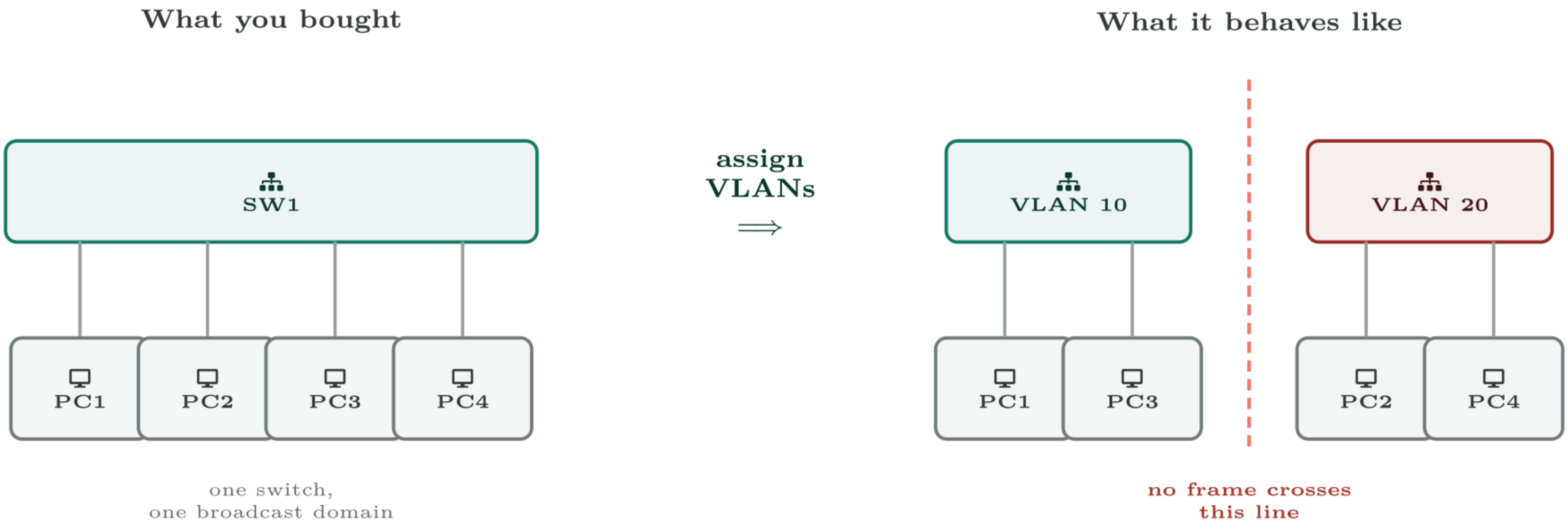
One line changes in the switching algorithm

Recall the algorithm from Chapter 3: learn the source, then forward, flood, or filter. VLANs change exactly one thing — **every step is now scoped to a VLAN**.

- Learning is per VLAN. The same MAC address may legitimately appear in two VLANs on the same switch.
- Flooding is per VLAN. An unknown unicast in VLAN 10 goes out every *VLAN 10* port and no others.
- A frame never leaves the VLAN it arrived in. Not once, not for broadcast, not for anything.

That is the whole mechanism. Everything else about VLANs — trunking, inter-VLAN routing, voice VLANs — is machinery for coping with the consequences of that one rule.

What a VLAN actually does



The whole idea, in one picture. Four ports on one switch become two broadcast domains that cannot reach each other — and nothing physical changed.

WATCH OUT

The consequence people forget

If two devices are in different VLANs they cannot communicate, *even if they are in the same IP subnet, plugged into adjacent ports on the same switch*. The switch will not carry the frame. Conversely two devices in the same VLAN but different subnets cannot communicate either, because IP will not deliver it.

VLAN and subnet are separate ideas that must be kept aligned by design. The convention that saves everyone trouble is one VLAN, one subnet, and a number that matches: VLAN 10 is `192.168.10.0/24`, VLAN 20 is `192.168.20.0/24`.

Creating VLANs and assigning ports

CONFIGURE

Three VLANs and the access ports that use them

IOS · configuration mode

```
! Create the VLANs and name them. A name is documentation that
! survives on the device -- use it.
vlan 10
 name Staff
vlan 20
 name Guest
vlan 30
 name Voice
exit
!
! Assign a single access port.
interface GigabitEthernet0/5
 description Staff desk 5
 switchport mode access
 switchport access vlan 10
 no shutdown
!
```

Three VLANs and the access ports that use them (continued)

IOS · configuration mode

```
! Assign a block of ports in one go.
interface range GigabitEthernet0/6 - 12
 description Staff desks 6-12
 switchport mode access
 switchport access vlan 10
!
interface range GigabitEthernet0/13 - 18
 description Guest desks
 switchport mode access
 switchport access vlan 20
```

CONCEPT

Why switchport mode access matters

Without it the port is left in dynamic auto mode, which means it will negotiate and may become a *trunk* if the device on the other end asks for one. On an access port that is both a fault waiting to happen and a security hole — an attacker who can negotiate a trunk can reach every VLAN.

Set it explicitly. It is one line, and it is the difference between a port that does what you configured and a port that does what the far end suggests.

Verifying, three ways

VERIFY

show vlan brief

SW1#

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/1, Gi0/2, Gi0/3, Gi0/4
10   Staff                            active    Gi0/5, Gi0/6, Gi0/7, Gi0/8
                                                Gi0/9, Gi0/10, Gi0/11, Gi0/12
20   Guest                            active    Gi0/13, Gi0/14, Gi0/15
30   Voice                            active
1002 fddi-default                     act/unsup
```

VERIFY

show interfaces GigabitEthernet0/5 switchport

SW1#

```
Name: Gi0/5
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Access Mode VLAN: 10 (Staff)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
```

show mac address-table vlan 10

SW1#

```
          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
  10    000a.f36e.7712    DYNAMIC     Gi0/5
  10    0060.5c2b.19a4    DYNAMIC     Gi0/7
```

Common pitfalls

- **Assigning a port to a VLAN that does not exist.** On many platforms the switch creates it silently; on others the port simply stops forwarding. Either way `show vlan brief` tells you within seconds.
- **Leaving everything in VLAN 1.** VLAN 1 cannot be deleted and carries control traffic. Do not put users on it, and do not use it as the management VLAN.
- **Forgetting that a VLAN can be shut down.** `shutdown` inside `vlan 10` stops the whole VLAN forwarding on that switch while every port still shows as correctly assigned. It is an unusual fault and therefore a nasty one.

Common pitfalls (continued)

- **Deleting a VLAN and leaving ports assigned to it.** Those ports become inactive. `show vlan brief` will not list them at all, which is confusing until you know to look for their absence.

A new desk cannot reach anything, and the port looks fine.

A user on `Gi0/9` has no connectivity. The port is up, the cable is good, and the configuration reads exactly like the working port next to it.

`Gi0/9` is not listed against VLAN 10 — nor against any other VLAN.

A new desk cannot reach anything, and the port looks fine.

What would you check first — and what do you expect to see?

show vlan brief | include Staff|Gi0/9

```
SW1#
10   Staff                                  active    Gi0/5, Gi0/6, Gi0/7, Gi0/8
```

show interfaces GigabitEthernet0/9 switchport

```
SW1#

Name: Gi0/9
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 15 (Inactive)
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. The port was assigned to VLAN 15, which does not exist on this switch — most likely a typing slip for 10, or a VLAN that was deleted after the port was configured. The word `(Inactive)` is the whole diagnosis: the port is administratively correct and operationally orphaned.

And the lesson

Fix. Either create VLAN 15 if it was meant to exist, or reassign the port: `switchport access vlan 10`. Confirm with `show vlan brief` that `Gi0/9` now appears in the Staff list, then confirm a MAC address learns against it.

Divide one switch into three networks

Topology. One switch, six PCs — two per VLAN — all in the same IP subnet to begin with.

1. With every port in VLAN 1, confirm all six PCs can ping each other. Record the MAC address table.
2. Create VLANs 10, 20 and 30 with names, and move two PCs into each.
3. Re-test. Which pings now succeed and which fail? Explain each result in terms of broadcast domains, not IP.
4. Run all three verification commands from this chapter and reconcile them.
5. Assign `Gi0/11` to VLAN 99, which you have not created. Record what `show vlan brief` and `show interfaces switchport` each say.

Divide one switch into three networks (continued)

6. Enter `vlan 20` and type `shutdown`. Confirm the ports still look correctly assigned while nothing passes. Undo it.
7. **Extension.** Re-address the PCs so each VLAN is its own subnet. Note that inter-VLAN pings still fail — and that fixing it needs Chapter 10.

Two PCs are in the same IP subnet, plugged into the same switch, in VLANs 10 and 20. Can they ping each other? Justify your answer at layer 2.

Discuss · then advance

Checkpoint 1 of 3

Two PCs are in the same IP subnet, plugged into the same switch, in VLANs 10 and 20. Can they ping each other? Justify your answer at layer 2.

> No. VLANs are separate broadcast domains, so the ARP request never reaches the other host. Sharing an IP subnet is irrelevant at layer 2 — and because both believe the destination is local, neither will send the traffic to a router either. This configuration cannot be fixed by routing; the VLANs must be corrected.

Which command distinguishes what you configured on a port from what the port has actually become?

Discuss · then advance

ANSWER

Checkpoint 2 of 3

Which command distinguishes what you configured on a port from what the port has actually become?

> `show interfaces <int> switchport`, which reports both `Administrative Mode` (what you configured) and `Operational Mode` (what it actually became).

A port shows `Access Mode VLAN: 15 (Inactive)`. State the two possible causes.

Discuss · then advance

ANSWER

Checkpoint 3 of 3

A port shows `Access Mode VLAN: 15 (Inactive)`. State the two possible causes.

> Either the VLAN does not exist in the VLAN database, or it exists and is shut down. `show vlan brief` distinguishes them.

What to take away

- A VLAN is a broadcast domain in software. Learning, flooding and forwarding all become per-VLAN operations.
- A frame never leaves its VLAN without being routed.
- Keep VLAN and subnet aligned one-to-one; mismatches produce faults that look like everything and nothing.
- Always set `switchport mode access` explicitly, or the port may negotiate a trunk.
- Verify three ways: `show vlan brief` by VLAN, `show interfaces switchport` by port, and the MAC table by evidence. Administrative versus operational mode is where faults show.

Where we got to

- A VLAN is a broadcast domain in software. Learning, flooding and forwarding all become per-VLAN operations.
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- Keep VLAN and subnet aligned one-to-one; mismatches produce faults that look like everything and nothing.
- Always set `switchport mode access` explicitly, or the port may negotiate a trunk.

NEXT

Chapter 9 — Edge-Host Connectivity: PoE, Voice, APs and IoT